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# **Analog Principles and Operation**

## **Programmable Logic Controllers**

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# Agenda

**Digital vs. Analog**

**Analog Types**

**Analog Devices**

**PID Control**

# Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

# Digital vs. Analog



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# Digital Defined

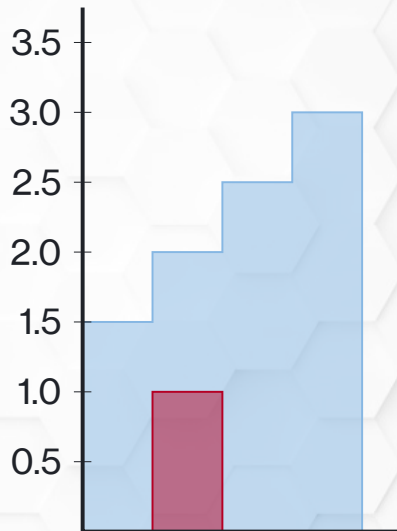
- Data or signals that have a fixed number of discrete values within their range
- Devices that use such data/signals



"Digital clock" by MichalPL, CC BY-SA 4.0, via [Wikimedia Commons](#)

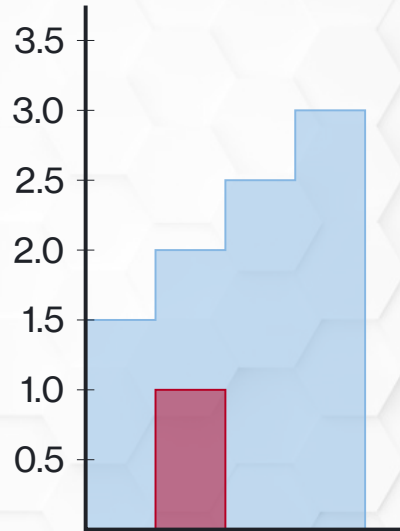
# Digital Signals (1 of 2)

- *Binary* is a special case of digital—the range is 0-1
- All binary signals are digital, not all digital signals are binary



# Digital Signals (2 of 2)

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



# Analog Defined

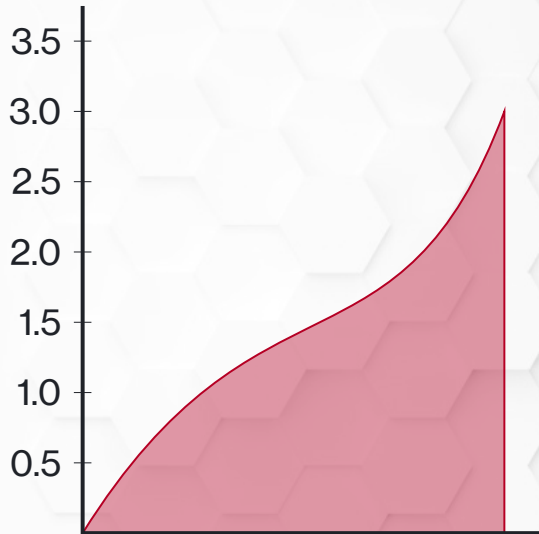
- Data or signals that vary continuously over a range of values
- Devices that use this type of data/signal



"Mercury-in-glass thermometer" by Anonimski, CC BY-SA 3.0, via [Wikimedia Commons](#)

# Analog Signals

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



# Why Analog?

- Motors use 69% of all electricity in the U.S. industrial sector<sup>1</sup>
- Motors running at full speed use more energy than motors running at reduced speed

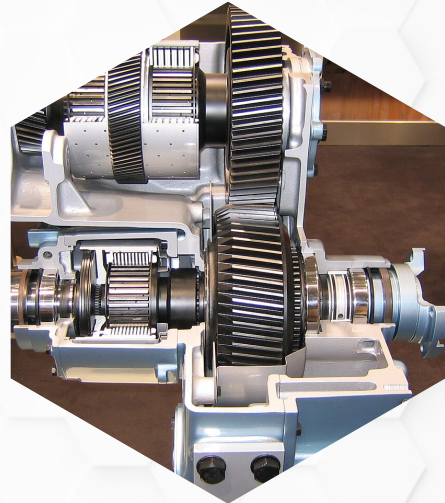


"Asynchronous electric motor" by Egzon123, CC BY-SA 3.0, via [Wikimedia Commons](#)

<sup>1</sup>Department of Energy, "Industrial Electric Motor Market Assessment" (2020).

# Controlling Motor Speed

- When a motor doesn't *need* to run at full speed you *could* use a gearbox...
- ...but that doesn't save any energy



"ZF gearbox at the Bauma trade fair 2004" by Aconcagua, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

# Saving Motor Energy

- Save energy (\$) by reducing motor speed
- VFDs *always* have an analog input
- Most have 2: voltage and current
- Can accept either potentiometer or PLC analog output



"Mitsubishi frequency converter type D700" by Ignac.mihevc at Slovenian Wikipedia, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

# Analog Types



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# Analog Signal Types

## Voltage

- **0 VDC-10 VDC**
- 0 VDC-5 VDC
- -10 VDC-10 VDC

## Current

- 0 mA-20 mA
- **4 mA-20 mA**

# Which Are the Most Common

Which types are the most common?

# Analog Example

- Two different analog sensors—identical except for their signal output
- Temperature range: 0°F-212°F
- One sensor produces a 0 VDC-10 VDC signal
- One sensor produces a 4 mA-20 mA signal

# Analog Scaling (1 of 2)

- Analog devices will produce an output that directly relates to the property being measured
- Almost always linear:  $1\text{ V} = 3^{\circ}\text{F}$ ,  $2\text{ V} = 6^{\circ}\text{F}$ ,  $3\text{ V} = 9^{\circ}\text{F}$ , etc.
- To convert divide the range of the sensor by the range of the analog signal

# Analog Scaling (2 of 2)

- $212^{\circ}\text{F} \div 10 \text{ VDC} = 21.2^{\circ}\text{F}/\text{V}$
- $212^{\circ}\text{F} \div 16 \text{ mA} = 13.25^{\circ}\text{F}/\text{mA}$ 
  - Remember, it starts at 4, the range is only 16
- Also remember that analog signals are *analog*—that means they're infinite
- These are values per unit, there an infinite number of values between

# Temperature Sensor (1 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| -42.40    | 0     | -26.5     | 4         |
| -21.20    | 0     | -13.5     | 4         |
| 0         | 0     | 0         | 4         |
| 21.2      | 1     | 13.25     | 5         |
| 31.8      | 1.5   | 53        | 8         |
| 106       | 5     | 106       | 12        |
| 159       | 7.5   | 159       | 16        |
| 201.4     | 9.5   | 198.75    | 19        |
| 212       | 10    | 212       | 20        |
| 233.2     | 10    | 225.25    | 20        |

# Temperature Sensor (2 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| -42.40    | 0     | -26.5     | 4         |
| -21.20    | 0     | -13.5     | 4         |
| 0         | 0     | 0         | 4         |

If the temperature decreases past the bottom of the sensor range the sensor will stay at the bottom value

# Temperature Sensor (3 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| 212       | 10    | 212       | 20        |
| 233.2     | 10    | 225.25    | 20        |

At the exact top of the sensor range the sensor will send out its maximum value

# Temperature Sensor (4 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| 212       | 10    | 212       | 20        |
| 233.2     | 10    | 225.25    | 20        |

The sensor will continue to send out its maximum value for any value above the sensor range

# Temperature Sensor (5 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| -42.40    | 0     | -26.5     | 4         |
| -21.20    | 0     | -13.5     | 4         |
| 0         | 0     | 0         | 4         |

Exactly *at* the bottom of the sensor range a 0 VDC-10 VDC sensor will register 0 VDC

# Temperature Sensor (6 of 6)

| Temp (°F) | Volts | Temp (°F) | Milliamps |
|-----------|-------|-----------|-----------|
| -42.40    | 0     | -26.5     | 4         |
| -21.20    | 0     | -13.5     | 4         |
| 0         | 0     | 0         | 4         |

Exactly *at* the bottom of the sensor range a 4 mA-20 mA sensor will register 4 mA

# Which to Pick

Which of these is preferred?

# Analog Signal Selection

## 0 VDC-10 VDC

- Easy unit conversion
- 0 is bottom, 10 is the top

## 4 mA-20 mA

- Current not as susceptible to noise as voltage
- Easy to diagnose—will *always* return at least 4 mA
- If you ever get a 0 something is wrong

# Analog Devices



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# Analog Inputs

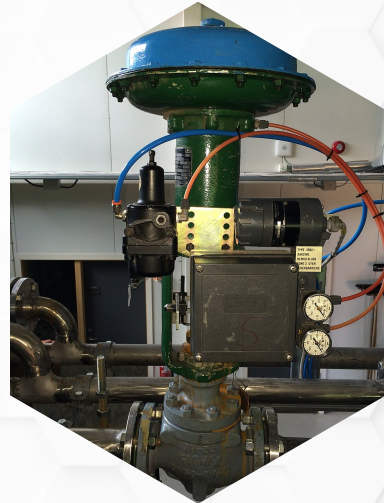
- Converts physical properties to analog signal
- A-to-D converter converts current/voltage to a binary number
- Some require external power source



"Pt-100 temperature sensor with cables" by Bitjungle, [CC BY-SA 4.0](#), via [Wikimedia Commons](#)

# Analog Outputs

- Converts analog signals to physical properties
- A-to-D converter converts analog signal to current/voltage
- Some require external power source



"A Fisher control valve with positioner" by Bitjungle, [CC BY-SA 4.0](#), via [Wikimedia Commons](#)

# Analog to Digital (A-to-D) Converter

- Analog *really* is infinite
- PLCs (and all computers) are digital—everything is 1's and 0's
- A-to-D converters convert the current/voltage to binary
- Each bit represents a certain number of volts/amps

# Maximum Analog Value (1 of 2)

- The maximum value of the binary number from the A-to-D depends on the *bit resolution* of the converter
- Let's take the example temperature sensors and plug them into a Micrologix 1000
- The Micrologix 1000 has a 10-bit analog resolution

# Maximum Analog Value (2 of 2)

| Bit       | 9    | 8   | 7   | 6   | 5  | 4  | 3  | 2 | 1 | 0 |
|-----------|------|-----|-----|-----|----|----|----|---|---|---|
| Bit Value | 512  | 256 | 128 | 64  | 32 | 16 | 8  | 4 | 2 | 1 |
| Total     | 1023 | 511 | 255 | 127 | 63 | 31 | 15 | 7 | 3 | 1 |

1023 is the largest number that can fit in 10 bits

# Analog Resolution

- The A-to-D has a bit resolution—the number of bits it uses to store the analog conversion
- This means that each bit correlates to a specific number of volts/amps
- This *also* means that each bit correlates to a specific real-world value

# Analog Resolution - Volts

- $10\text{ V} \div 1023 = 0.009775\text{ V/bit}$
- $21.2^{\circ}\text{F/V} \times 0.009775\text{ V/bit} = 0.20723^{\circ}\text{F/bit}$
- This is the *resolution*—this PLC can't see increments smaller than  $0.20723^{\circ}\text{F}$

| Bit   | 9 | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1     | 0      |
|-------|---|-----|-------|--------|--------|--------|--------|--------|-------|--------|
| Volts | 5 | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019 | 0.0098 |

Note: table values have been heavily rounded for formatting

# Analog Resolution - mA

- $20\text{ V} \div 1023 = 0.01955\text{ mA/bit}$
- Note: this time we need the *max*—we use 20, not 16
- $13.25^{\circ}\text{F/mA} \times 0.01955\text{ mA/bit} = 0.259^{\circ}\text{F/bit}$
- This is the *resolution*—this PLC can't see increments smaller than  $0.259^{\circ}\text{F}$

| Bit | 9  | 8 | 7     | 6     | 5      | 4      | 3      | 2      | 1      | 0     |
|-----|----|---|-------|-------|--------|--------|--------|--------|--------|-------|
| mA  | 10 | 5 | 2.502 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019 |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 1 (1 of 2)

What 10-bit number represents 106°F?

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 1 (2 of 2)

- we know that 106°F is half the scale—5 V/12 mA
- Start adding up bits to get to those values and do the binary conversion
- 0-10 V sensor: 512
- 4-20 mA sensor: 614

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 2 (1 of 2)

What 10-bit number represents 72°F?

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 2 (2 of 2)

$$72^{\circ}\text{F} \div 212^{\circ}\text{F} = 0.34$$

$$0.34 \times 10 \text{ V} = 3.4 \text{ V}$$

$$3.4 \text{ V} = 342$$

$$0.34 \times 16 \text{ mA} = 5.44 \text{ mA}$$

$$5.44 \text{ mA} + 4 \text{ mA} = 9.44 \text{ mA}$$

$$9.44 \text{ mA} = 481$$

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 3 (1 of 2)

What 10-bit number represents 0°F?

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Temperature to Analog 3 (2 of 2)

For 0-10V, 0 is 0

0°F = 4 mA  
4 mA = 204

| Bit       | 9   | 8   | 7     | 6      | 5      | 4      | 3      | 2      | 1      | 0      |
|-----------|-----|-----|-------|--------|--------|--------|--------|--------|--------|--------|
| Bit Value | 512 | 256 | 128   | 64     | 32     | 16     | 8      | 4      | 2      | 1      |
| Volts     | 5   | 2.5 | 1.251 | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  | 0.0098 |
| mA        | 10  | 5   | 2.502 | 1.251  | 0.6256 | 0.3128 | 0.1564 | 0.0782 | 0.0391 | 0.019  |

Note: table values have been heavily rounded for formatting

# Application

- In the real world that math almost *never* works
- These are all physical devices, no physical device is perfect
- It will get you close but usually not close enough

# Analog Devices

## Analog Scaling



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# Analog Scaling

- Analog inputs come in as integers—whole numbers that represent a voltage or current value from the sensor
- Most analog cards are 12-bit—the value received by the PLC will be 0-4096
- You *can* use this value, but it doesn't mean much to anyone else looking at the code
- It is *much* better to scale it to engineering units *before* using it in logic

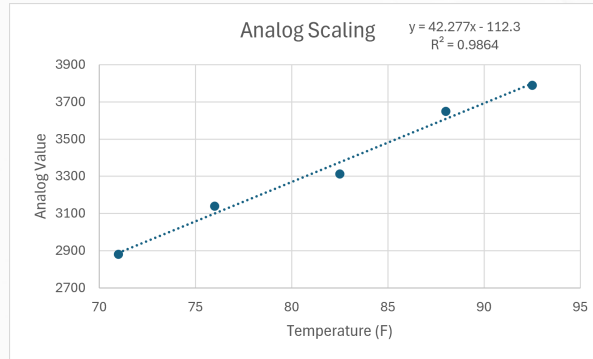
# Allen-Bradley Scaling

- Allen-Bradley's preferred method in the Logix 5000 PLCs is to scale the value in the definition of the analog card itself
- Not all cards and input devices are equal—if you swap temperature sensors and it's off compared to the old one the scaling needs to be changed
- If the scaling is in the card definition, the system needs to be taken offline to change it
- Plus, not all manufacturers do it this way—we like “hardware agnostic” programming methods

# Setting the Scale (1 of 2)

- Math would tell us the scaling factor would be:  
$$\text{Engineering Units} = \text{UnitRange} \times \frac{\text{AnalogValue}}{\text{MaxValue}}$$
- This only works *if* the signal starts at 0
- It is also only ever going to get you close—not all sensors and cards are equal
- Best practice is to actually *measure* the conversion factor
- In this case, a spreadsheet is your best friend...

# Setting the Scale (2 of 2)



Set the sensor to 5-6 known real-world values and read the analog input value. Plot them in a spreadsheet and run a *linear regression*—which the software will do automatically with one click. The regression equation is the scaling factor.

# PID Control



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# Questions?

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